Ericsson BTM-PE-05 Hardware/Environmental Fault (ASIC Memory Parity Errors, Line Card Crashes, Power Supply Redundancy Loss) Troubleshooting and Resolution SOP

Node: BTM-PE-05 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_1293

Description

This Standard Operating Procedure outlines the necessary diagnostic steps to identify and resolve hardware/environmental faults in the BTM-PE-05 node at the BTM Layout Aggregation site, running Ericsson vendor equipment. The PE role and specific failure domain are characterized by ASIC memory parity errors, line card crashes, and power supply redundancy loss. The BTM-PE-05 node is a critical component in the Ericsson network architecture, operating as a Provider Edge (PE) device, facilitating connectivity between the provider and customer networks. The failure of this node can result in service disruptions and data loss. To address this failure, the network administrator must perform a series of diagnostic and resolution steps, leveraging Ericsson-specific CLI commands to identify the root cause of the issue and implement corrective measures.

Pre-requisite Checks

- Verify redundant link status
- Ensure remote hands are dispatched to the BTM Layout Aggregation site
- Conduct a site survey to verify environmental conditions

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy status: active/active, Description: Verify power supply redundancy status using the show system redundancy command. This command will display the current redundancy status, indicating whether the node is operating in active/active mode.

Command: show system asic memory, Expected Output: ASIC memory status: healthy, Description: Use the show system asic memory command to verify the status of the ASIC memory. This command will display the
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current memory status, indicating whether the memory is healthy or
not.

Command: show system linecard, Expected Output: Line card status:
online, Description: Use the show system linecard command to verify
the status of the line card. This command will display the current
line card status, indicating whether the line card is online or not.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system asic, Description: If the ASIC memory parity errors are identified as the root cause, replace the faulty ASIC using the replace system asic command. This command will initiate the replacement process, ensuring the node remains operational.

Step 2: Step: Replace the faulty line card, Command: replace system linecard, Description: If the line card crashes are identified as the root cause, replace the faulty line card using the replace system linecard command. This command will initiate the replacement process, ensuring the node remains operational.

Step 3: Step: Replace the faulty power supply, Command: replace system power-supply, Description: If the power supply redundancy loss is identified as the root cause, replace the faulty power supply using the replace system power-supply command. This command will initiate the replacement process, ensuring the node remains operational.

Post-Check Verification

```
Command: show system redundancy, Expected Output: Redundancy status:
active/active, Description: Verify power supply redundancy status
using the show system redundancy command. This command will display
the current redundancy status, indicating whether the node is
operating in active/active mode.

Command: show system asic memory, Expected Output: ASIC memory status:
healthy, Description: Use the show system asic memory command to
verify the status of the ASIC memory. This command will display the
current memory status, indicating whether the memory is healthy or
not.

Command: show system linecard, Expected Output: Line card status:
online, Description: Use the show system linecard command to verify
the status of the line card. This command will display the current
line card status, indicating whether the line card is online or not.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: config restore, Description: If the resolution steps fail to resolve the issue, restore the previous configuration using the config restore command. This command will revert the node to its previous state, ensuring minimal disruption to the network.

- Step: Revert to previous software version, Command: software revert, Description: If the resolution steps fail to resolve the issue, revert to the previous software version using the software revert command. This command will roll back the node to its previous software version, ensuring minimal disruption to the network.